

ABC BANK

FI Agent — Personal Loan

Customer Journey · System Architecture · Data Capture Guide

Prepared: 30 May 2026 23:52

1. Purpose & Scope

System	ABC Bank Field Investigation (FI) Agent — Personal Loan origination
Purpose	Automate identity verification, property documentation, and income assessment for personal loan applications via a mobile-friendly web interface
Audience	Customer (loan applicant), Field Investigation Officer, Technology Team
Version	2.0 30 May 2026 23:52

2. System Architecture

Components

Web Frontend	Vite + TypeScript SPA served at /fi/ — handles camera, audio, geolocation, photo capture, blur detection, bank statement upload, and session recording
WebSocket Server	FastAPI + Python — drives the entire FI session in real-time; sends questions, TTS audio, photo prompts; receives answers and photo ACKs
REST API	FastAPI — receives photos (immediate), bank statement PDF, and final submit (metadata + WebM recording); triggers background report generation
AWS Polly	Neural TTS (Kajal en-IN voice) — converts each question to MP3 audio played on the applicant's device
AWS Transcribe	Streaming speech-to-text — applicant's spoken answers transcribed in real-time over the WebSocket stream

AWS Textract	OCR engine — extracts text from PAN card photo and home nameplate
AWS Rekognition	Face match — compares selfie photo with PAN card face to verify identity
OpenAI GPT-4o	Analyses property images and bank statement; produces structured credit assessment
Google Maps API	Reverse geocodes the session centroid to a human-readable address
ReportLab	Generates the 10-section PDF credit report server-side
Storage (disk)	Case folder per session: photos, recording, metadata, analysis JSON, PDF

Request Flow (simplified)



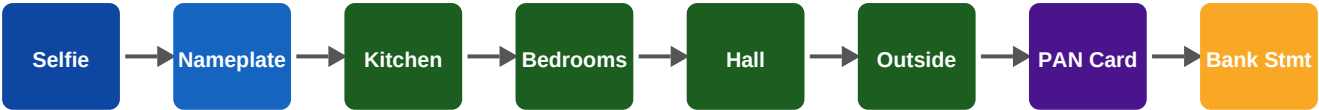
3. Customer Journey



Step 1 — Marketing Page	/fi/ → ABC Bank landing page with loan benefits, interest rates, and 'Apply Now' CTA
Step 2 — Application Form	Customer enters: First Name, Last Name, Date of Birth, Address, City, PAN Number, Mobile Number, Annual Income Range
Step 3 — Review	All form data shown in a summary table; customer confirms before starting the FI process
Step 4 — FI Session	Automated session: 3 verbal Q&A; → selfie → 7 home photos (nameplate first) → PAN card → bank statement upload Session recorded throughout as WebM video+audio
Step 5 — Submit	'Submit FI Report' button uploads session recording + metadata; background report generation starts
Step 6 — Thank You	Full-screen confirmation: case reference number + registered mobile number displayed; customer informed that ABC Bank will contact them on their mobile

4. FI Session — Detailed Flow

Photo & Document Capture Sequence



Q&A; Phase (3 questions, spoken & transcribed)

- Q1: Please confirm your full name
- Q2: What is your residential PIN code?
- Q3: What is the name of your city?
- GPS coordinates captured at start of each answer
- Answer shown on screen for confirmation; Re-record option available
- Server retries if answer is empty or inaudible

Photo Phase — per-photo flow

- Server sends 'capture_photo' → countdown timer (5 sec configurable)
- Photo captured automatically from camera
- Blur detection (Laplacian variance): score < 40 → auto-retake (up to 2 times)
- Review screen: Save (5-second auto-save) or Discard (retake)
- Photo uploaded immediately to server via REST; GPS + blur score recorded
- PAN card: up to 3 OCR retries if text not readable

Bank Statement Upload

- Server requests PDF upload after all photos
- Customer taps 'Upload PDF' → OS file picker
- PDF uploaded via REST; server saves to session folder
- Skip option available (income section shown as INCOMPLETE in report)

5. Data Captured Per Session

Application Form	First/Last Name, DOB, Address, City, PAN, Mobile, Income Range
Q&A; Answers	Text transcript + GPS coordinates (lat/lon/timestamp) per answer
Photos (up to 9)	JPEG files + GPS + blur score + OCR text (nameplate/PAN)
PAN Verification	Textextract OCR (PAN no., name, father, DOB) + NSDL result + Rekognition face-match score vs selfie
3-Way Name Match	Application form name ↔ PAN OCR name ↔ Nameplate OCR text
Bank Statement	PDF file + GPT-4 income analysis JSON
GPS Trail	All lat/lon readings — geo-verified (max spread ≤ 500m = PASS)
Session Recording	WebM video+audio of the entire session (browser captureStream)
Metadata JSON	session_data.json — denormalised single source of truth

6. WebSocket Message Protocol

Server → Client

{type:'question', index, text}	Ask next interview question (TTS MP3 embedded)
{type:'start_listening', timeout_ms}	Begin STT recording
{type:'transcript', text, is_final}	Partial or final transcript
{type:'announce_photo', prompt}	Announce upcoming photo
{type:'countdown', value}	Tick-by-tick countdown (5...1)
{type:'capture_photo', index}	Trigger camera capture
{type:'pan_retry', attempt, message}	PAN OCR failed — retake requested
{type:'request_document', prompt}	Request bank statement upload
{type:'session_done', message}	All steps complete — show Submit button

Client → Server

{type:'ready', session_id, basic_info}	Handshake after WS open
Binary PCM frames	16 kHz / 16-bit mono audio stream for Transcribe
{type:'audio_end'}	STT recording finished
{type:'answer_confirm'} / {type:'answer_retry'}	Confirm transcript or request re-record
{type:'tts_done'}	Device finished playing TTS audio
{type:'photo_taken', filename}	Photo uploaded; acknowledge server
{type:'document_ready', filename}	Bank statement uploaded

7. Post-Submission Report Generation



Sec 1 — Application Details	Basic info from form
Sec 2 — Executive Summary	Session KPIs, risk preview
Sec 3 — Field Interview	Q&A; table with GPS per answer
Sec 4 — Location Verification	GPS spread, geocoded address, PASS/FAIL
Sec 5 — Home Evidence	Photos + analysis + blur + nameplate OCR
Sec 6 — PAN Verification	OCR, 3-way name match, NSDL, face match

Sec 7 — Income Analysis	Bank statement metrics + risk/positive flags
Sec 8 — Credit Assessment	Score + grade + 4-pillar breakdown + conditions
Sec 9 — Scoring & Risk	Rule-based cross-check score
Sec 10 — Recommendations	Findings, final decision box, signature block

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